## Task 1: Introduction to Numpy package

### Task 1_1

Run Task1_1

```
C:\Users\casey\PycharmProjects\PythonProject\.venv\Scrip
Array: [10 11 12 13 14 15 16 17 18 19]
Shape: (10,)
Dimensions: 1
Data Type: int64

Process finished with exit code 0
```

Task1_1.py

```python
import numpy as np

arr = np.arange( start_or_stop: 10, stop: 20)
print("Array:" , arr)
print("Shape:", arr.shape)
print("Dimensions:" , arr.ndim)
print("Data Type:", arr.dtype)
```

### Task 1_2

Run Task1_2

```
C:\Users\casey\PycharmProjects\PythonProject\.venv\Scrip
[[0 0 0]
 [0 7 0]
 [0 0 0]]

Process finished with exit code 0
```

Task1_1.py Task1_2.py

```python
import numpy as np

arr = np.zeros( shape: (3, 3), dtype=int)
Tab to complete  Full Line completion runs locally
arr[1, 1] = 8  Ctrl+Right to complete word
print(arr)
arr[1, 1] = 9
print(arr)
```

### Task 1_3

Run Task1_3

```
C:\Users\casey\PycharmProjects\PythonProject\.venv\Scrip
Second row: [40 50 60]
Third column: [30 60 90]
Submatrix (first 2 rows and cols) :
[[10 20]
 [40 50]]

Process finished with exit code 0
```

Task1_1.py Task1_2.py Task1_3.py

```python
import numpy as np

arr = np.array([[10, 20, 30],
                [40, 50, 60],
                [70, 80, 90]])

print("Second row:", arr[1])
print("Third column:", arr[:, 2])
print("Submatrix (first 2 rows and cols) :")
print(arr[:2, :2])
```

### Task 1_4

Run Task 1_4

```
C:\Users\casey\PycharmProjects\PythonProject\.venv\Scrip
[[[1 1 1]
  [1 1 1]]

 [[1 1 5]
  [1 1 1]]]

Process finished with exit code 0
```

Task1_1.py Task1_2.py Task1_3.py Task 1_4.py

```python
import numpy as np
arr = np.ones( shape: (2, 2, 3), dtype=int)
arr[1, 0, 2] = 5
print(arr)
```

## Task 1_5

```
Run  Task 1_5
C:\Users\casey\PycharmProjects\PythonProject\.venv\Scri
[ 2  6 10]

Process finished with exit code 0
```

```
Task1_1.py  Task1_2.py  Task1_3.py  Task 1_4.py  Task 1_5.py
import numpy as np

arr = np.array([2, 4, 6, 8, 10])
print(arr[::2])
```

## Task 1_6

```
Run  Task 1_6
C:\Users\casey\PycharmProjects\PythonProject\.venv\Scri
Sum: [ 7 14 21]
Difference: [3 6 9]
Product: [10 40 90]
Divisions: [2 2 2]

Process finished with exit code 0
```

```
Task1_1.py  Task1_2.py  Task1_3.py
import numpy as np

a = np.array([5, 10, 15])
b = np.array([2, 4, 6,])
print("Sum:", a+b)
print("Difference:", a-b)
print("Product:", a*b)
print("Divisions:", a//b)
print("Modulo:", a%b)  Tab to complete
```

## Task 1_7

```
Run  Task 1_7
C:\Users\casey\PycharmProjects\PythonProject\.venv\Scri
[[ 1  0 -3]
 [ 4  0 -6]
 [ 7  0 -9]]

Process finished with exit code 0
```

```
Task1_1.py  Task1_2.py  Task1_3.py  Task 1_7.py
import numpy as np

matrix = np.arange( start_or_stop: 1,  stop: 10).reshape(3, 3)
vector = np.array([1, 0, -1])

result = matrix*vector
print(result)
```

## Task 1_8

```
Task1_1.py  Task1_2.py  Task1_3.py  Task 1_7
import numpy as np

arr = np.arange( start_or_stop: 0,  stop: 16,  step: 3)
sqrt_arr = np.sqrt(arr)
print(sqrt_arr)
```

```
Run  Task 1_8
C:\Users\casey\PycharmProjects\PythonProject\.venv\Scripts\python.ex
[0.         1.73205081 2.44948974 3.         3.46410162 3.87298335]

Process finished with exit code 0
```

## Task 1_9

**INCOMPLETE, COME BACK**

### Task 1_10

```
Run  Task 1_10
C:\Users\casey\PycharmProjects\PythonProject\.venv\Scrip
[[28.  35.  42. ]
 [ 4.   7.   5. ]
 [15.  20.  27.5]]
[[25 30 35]
 [26 31 36]]

Process finished with exit code 0
```

```
Task 1_9.py   Task 1_10.py
import numpy as np

growth_factors = np.array([0.7, 0.2, 0.5])
rainfall = np.array([[40, 50, 60],
                     [20, 35, 25],
                     [30, 40, 55]])

temps = np.array([
    [[10, 15, 20], [25, 30, 35]],
    [[11, 16, 21], [26, 31, 36]]
])

height_increase = growth_factors[:, np.newaxis] * rainfall
print(height_increase)
print(temps[:, 1, :])
```

### Task 2: Introduction to OpenCV package

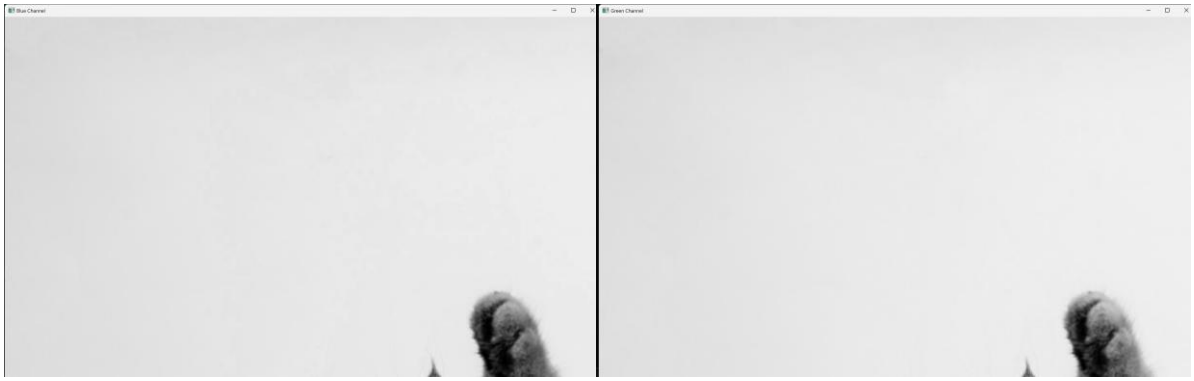
I will be using “cat.jpg” titled as “image.png” for these tasks. I could not find a zipped folder for week 5.

### Task 2_1

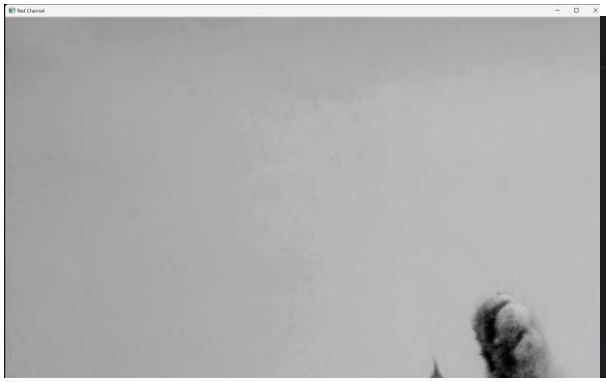
```
Run  Task 2_1
C:\Users\casey\PycharmProjects\PythonProject\.venv\Scrip
Type of the image: <class 'numpy.ndarray'>
Shape of the image array: (2879, 1920, 3)

Process finished with exit code 0
```

```
Task 1_10.py   Task 2_1.py
import cv2

image = cv2.imread('image.png')
print(f"Type of the image: {type(image)}")
print(f"Shape of the image array: {image.shape}")
```

### Task 2_2



Blue Channel, Green Channel

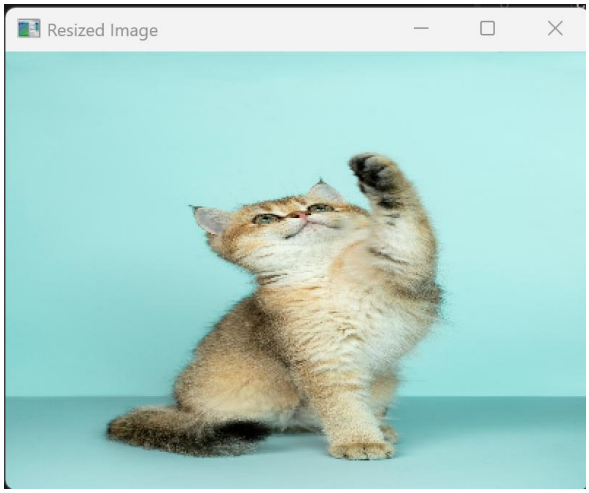


Task 1_10.py  Task 2_1.py  Task 2_2.py

```python
import cv2

image= cv2.imread('image.png')
b, g, r = cv2.split(image)
cv2.imshow( winname: 'Blue Channel', b)
cv2.imshow( winname: 'Green Channel', g)
cv2.imshow( winname: 'Red Channel', r)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Red Channel

**Task 2_3**




Task 1_10.py  Task 2_1.py  Task 2_2.py  Task

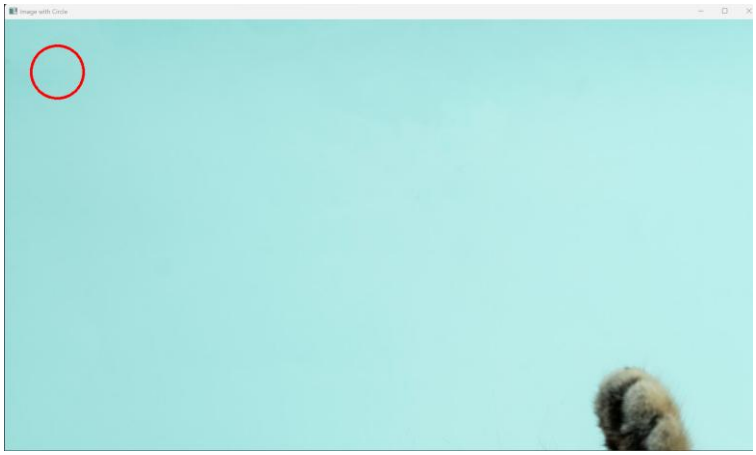
```python
import cv2

image = cv2.imread('image.png')
resized_img = cv2.resize(image,  dsize: (400, 300))
cv2.imshow( winname: 'Resized Image', resized_img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

**Task 2_4**

Task 1_10.py  Task 2_1.py  Task 2_2.py  Task 2_3

```python
import cv2

image = cv2.imread('image.png')
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
cv2.imwrite( filename: 'gray_image.png', gray_image)
```

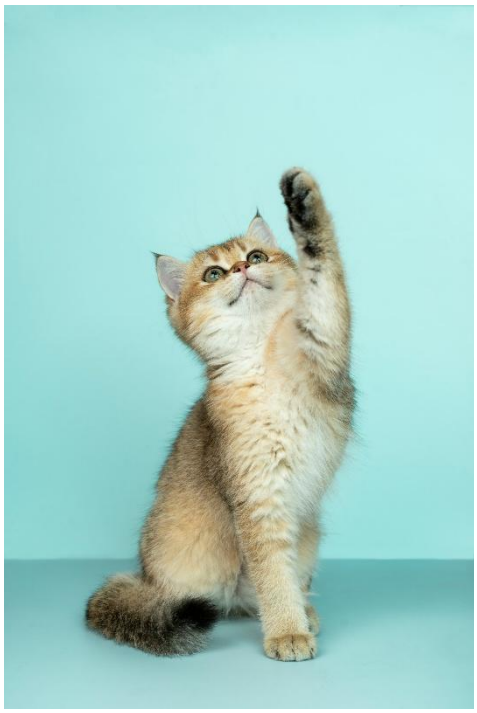
## Task 2_5



Task 1_10.py | Task 2_1.py | Task 2_2.py | Task 2_3.py | Task 2_4.py | Task 2_5.py

```python
import cv2

image = cv2.imread('image.png')
image_copy = image.copy()
cv2.circle(image_copy, center: (100, 100), radius: 50, color: (0, 0, 255), thickness: 3) # BGR: Red
cv2.imshow( winname: 'Image with Circle', image_copy)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

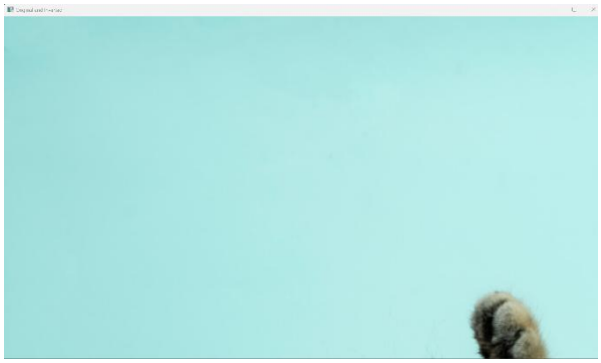
## Task 2_6



Task 2_1.py | Task 2_2.py | Task 2_3.py | Task 2_4.py

```python
import cv2

image = cv2.imread('image.png')
print(f"Original pixel value at (50, 50): {image[50, 50]}")
image[50, 50] = [255, 255, 255]
cv2.imwrite( filename: 'modified.png', image)
```

## Task 2_7

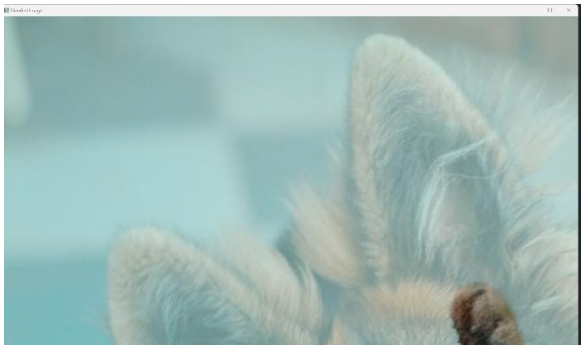


Task 1_10.py  Task 2_1.py  Task 2_6.py  Task 2_7.py

```python
import cv2
import numpy as np

image = cv2.imread('image.png')
inverted = 255 - image

combined = np.hstack((image, inverted))
cv2.imshow( winname: 'Original and Inverted', combined)
cv2.waitKey(0)
```

## Task 2_8

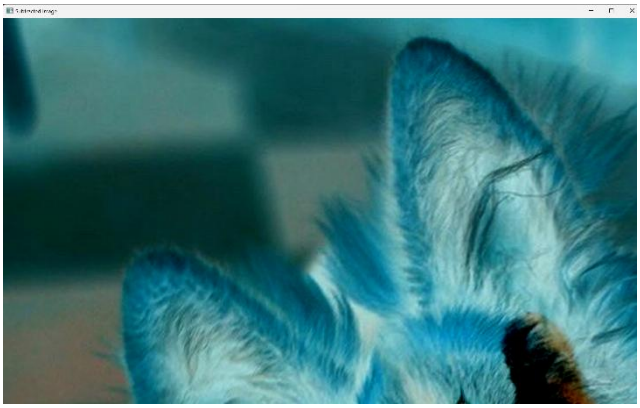
I have used image “dog

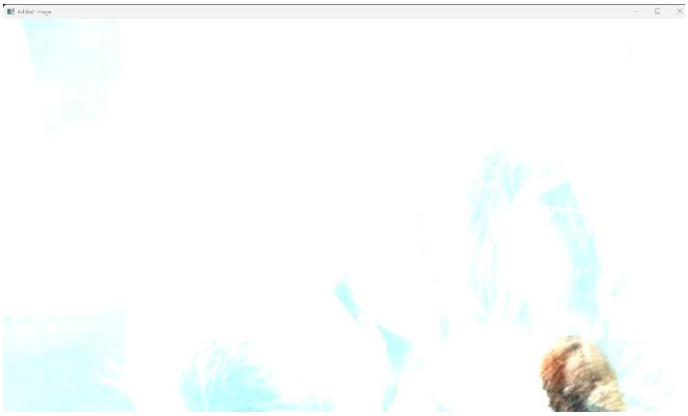


Task 1_10.py  Task 2_1.py  Task 2_6.py  Task 2_7.py  Task 2_8.py

```python
import cv2

img1 = cv2.imread('image.png')
img2 = cv2.imread('image2.png')

img2_resized = cv2.resize(img2,  dsize: (img1.shape[1], img1.shape[0]))
blended = cv2.addWeighted(img1,  alpha: 0.6, img2_resized,  beta: 0.4,  gamma: 0)
cv2.imshow( winname: 'Blended Image', blended)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

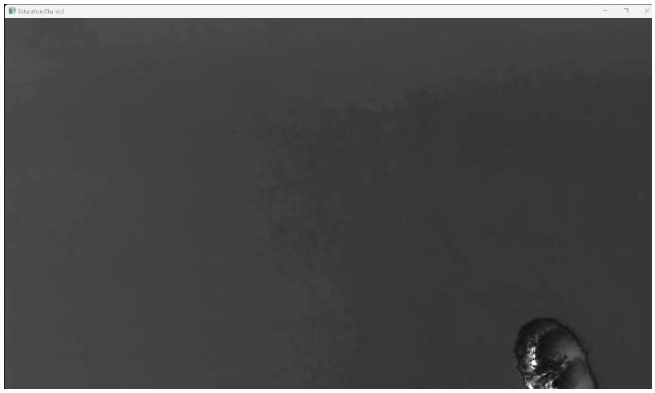
## Task 2_9



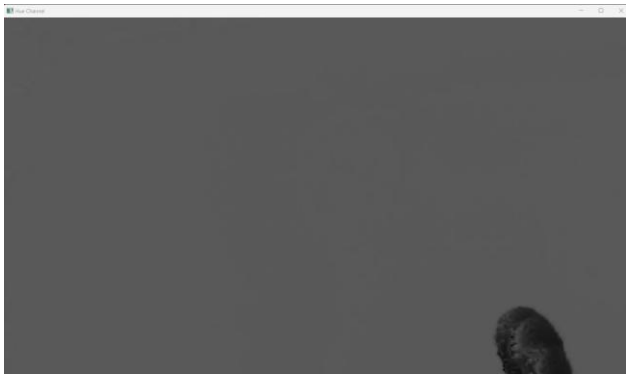


```python
import cv2

img1 = cv2.imread('image.png')
img2 = cv2.imread('image2.png')
img2_resized = cv2.resize(img2,  dsize: (img1.shape[1], img1.shape[0]))

added = cv2.add(img1, img2_resized)
subtracted = cv2.subtract(img1, img2_resized)

cv2.imshow( winname: 'Added Image', added)
cv2.imshow( winname: 'Subtracted Image', subtracted)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

**Task 2_10**



Value, Saturation



```python
import cv2

image = cv2.imread('image.png')
hsv_img = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)
h, s, v = cv2.split(hsv_img)

cv2.imshow( winname: 'Hue Channel', h)
cv2.imshow( winname: 'Saturation Channel', s)
cv2.imshow( winname: 'Value Channel', v)
cv2.waitKey(0)
```

Hue